Exp 18: EXPERIMENT TO UNDERSTAND KNN ALGORITHM FOR A GIVEN DATA SET.

AIM: Understand the KNN algorithm for the dataset given.

SAMPLE CODE:

```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, classification_report

# Load the dataset
df = pd.read_csv('Iris.csv')

# Display dataset information
df.info()

# Display the count of each flower variety
print(df['variety'].value_counts())

# Display the first five rows
print(df.head())
```

```
# Separate features and labels
features = df.iloc[:, :-1].values
label = df.iloc[:, 4].values

# Split data into training and testing sets
xtrain, xtest, ytrain, ytest = train_test_split(
   features, label, test_size=0.2, random_state=0
)

# Create and train the KNN model
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)

# Calculate training and testing accuracy
print("Training Accuracy:", model_KNN.score(xtrain, ytrain))
print("Testing Accuracy:", model_KNN.score(xtest, ytest))

# Predict flower varieties for the full dataset
predictions = model_KNN.predict(features)

# Display confusion matrix
print("Confusion Matrix:")
print(confusion_matrix(label, predictions))

# Display classification report
print("Classification Report:")
print(classification_report(label, predictions))
```

OUTPUT:

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.variety.value_counts()
```

```
Setosa        50
Versicolor    50
Virginica     50
Name: variety, dtype: int64
```

```
df.head()
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
```

```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

```
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,randc
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

```
KNeighborsClassifier()
```

```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
```

```
0.9583333333333334
1.0
```

```
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
```

```
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

```
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

RESULT: Thus, the experiment to understand KNN algorithm for a given data is conducted.